

Tyler Trotter

Exam 4 Foundations of Analysis I Math 3210
• I will not use any outside resource on this exam.
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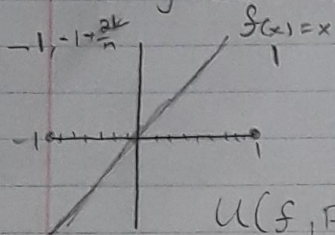
Problem #1.

Using the def. of Integrability (or an equivalent integrability criteria) prove that the function $f(x) = x$ is integrable on $[-1, 1]$ and that $\int_{-1}^1 x dx = 0$.

A function is integrable if there exists a partition P such that

$$\lim_{n \rightarrow \infty} (U(f, P_n) - L(f, P_n)) = 0.$$

Since $f(x) = x$ is monotone on the interval and continuous, it is integrable. Proceed by computing the upper and lower sums.



Consider a partition which divides the interval $[-1, 1]$ into n equal lengths or sub-intervals. The upper and lower sums are:

$$U(f, P_n) = \sum_{k=1}^n M_k (x_k - x_{k-1})$$

$\sup_{x \in [x_{k-1}, x_k]} f(x)$

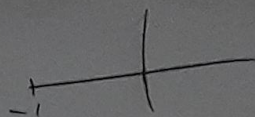
$$L(f, P_n) = \sum_{k=1}^n m_k (x_k - x_{k-1})$$

$\inf_{x \in [x_{k-1}, x_k]} f(x)$

We want to show $\lim (U(f, P_n) - L(f, P_n)) = 0$.

cont. on next page.

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$



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Problem #1, continued.

Split the interval into two equal lengths, both containing 0.

$[-1, 0]$ and $[0, 1]$.

For the second interval, choose a partition such that we divide $[0, 1]$ into $\frac{1}{n}$ equal lengths.

$$U(f, P_n) = \sum_{k=1}^n (f(k) \cdot (\frac{1}{n})) = \sum_{k=1}^n \frac{k}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=1}^n k$$

Using the given formula, we get $\frac{1}{n^2} \sum_{k=1}^n k = \frac{n(n+1)}{2n^2}$

$$L(f, P_n) = \sum_{k=1}^n (f(k-1) \cdot (\frac{1}{n})) = \sum_{k=1}^n \frac{k-1}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=1}^n (k-1) = \frac{n(n-1)}{2n^2}$$

$$\lim (U(f, P_n) - L(f, P_n)) = \lim \left(\frac{n(n+1) - n(n-1)}{2n^2} \right)$$

$$= \lim \frac{2n}{2n^2} = \lim \frac{1}{n} = 0.$$

For the other interval, $[-1, 0]$, we also divide into $\frac{1}{n}$ equal length subintervals for our partition.

$$U(f, P_n) = \sum_{k=1}^n f(k) \left(\frac{1}{n} \right) = \sum_{k=1}^n \left(-\frac{k}{n} \right) \left(\frac{1}{n} \right) = -\frac{1}{n^2} \sum_{k=1}^n k = -\frac{n(n+1)}{2n^2}$$

$$L(f, P_n) = \sum_{k=1}^n f(k-1) \left(\frac{1}{n} \right) = \sum_{k=1}^n \left(-\frac{k-1}{n} \right) \left(\frac{1}{n} \right) = -\frac{1}{n^2} \sum_{k=1}^n (k-1) = -\frac{n(n-1)}{2n^2}$$

$$\lim (U(f, P_n) - L(f, P_n)) = \lim \left(\frac{-n(n+1) - (-n(n-1))}{2n^2} \right) = \lim \frac{-2n}{2n^2} = 0.$$

So, $f(x) = x$ is integrable on the entire interval. Now we can use any sum and add the intervals up. We get

$$\int_0^1 x dx = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} \left(\frac{n(n+1)}{2n^2} \right) = \frac{1}{2}$$

$$+ \int_{-1}^0 x dx = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} \left(-\frac{n(n+1)}{2n^2} \right) = -\frac{1}{2} \quad \left[\frac{1}{2} + \left(-\frac{1}{2} \right) = 0 \right]$$

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Problem #2:

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Prove that $\lim_{n \rightarrow \infty} \int_{1/2}^1 \cos^n(x) dx = 0$

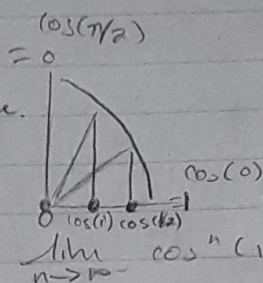
$$\int_a^b \lim_{n \rightarrow \infty} f_n(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$$

We proved in the homework that we could bring limits in/out of integrals without impacting the result.

So, I get $\int_{1/2}^1 \left[\lim_{n \rightarrow \infty} \cos^n(x) \right] dx$

Consider the unit circle.

$$0 < \cos(1) < \cos(1/2) < 1$$



Then we consider $\lim_{n \rightarrow \infty} \cos^n(1)$ and $\lim_{n \rightarrow \infty} \cos^n(1/2)$

Since $0 < \cos(1) < 1$ and $0 < \cos(1/2) < 1$, we have a positive fraction, for each of $\cos(1)$ and $\cos(1/2)$. Due to this, I am able to assert that $\cos^n(1) = [\cos(1)]^n$ and $\lim_{n \rightarrow \infty} [\cos(1)]^n \rightarrow 0$ because $\cos(1)$ or $\cos(1/2)$ are represented as follows $\left(\frac{a}{b}\right)$ where $a < b$. This means $a < b$. Then $\lim_{n \rightarrow \infty} \left(\frac{a}{b}\right)^n = \lim_{n \rightarrow \infty} \frac{a^n}{b^n}$ which goes to 0 because $a < b$ ($a^k < b^k$ means b^k grows much faster). Now, for any $x \in (1/2, 1)$, $\lim_{n \rightarrow \infty} \cos^n(x) \rightarrow 0$. So, we have $\int_{1/2}^1 0 dx = 0 \cdot \left. 1 \right|_{1/2}^1 = 0$.

Problem #3:

(Consider the power series: $\sum_{k=1}^{\infty} \frac{(x-5)^k}{3^k k}$)

a) Find the interval of convergence for this series.

Use Ratio Test:
$$r = \lim \left| \frac{(x-5)^k (x-5)}{3 \cdot 3^k (k+1)} \cdot \frac{3^k \cdot k}{(x-5)^k} \right|$$

Continue here:
 $x \in (2, 8)$; check endpoints:

Check
 $x=2$:

$$\sum_{k=1}^{\infty} \frac{(2-5)^k}{3^k k} = \sum_{k=1}^{\infty} \frac{(-3)^k}{3^k k} = \sum_{k=1}^{\infty} \frac{(-1)^k}{k}$$

which converges

conditionally (alternating harmonic)

Check

$$x=8: \sum_{k=1}^{\infty} \frac{(8-5)^k}{3^k k} = \sum_{k=1}^{\infty} \frac{3^k}{3^k k} = \sum_{k=1}^{\infty} \frac{1}{k} \text{ which diverges (harmonic).}$$

So, the interval of convergence is: $x \in [2, 8)$

b) If we define $f(x)$ to be the sum of the series $\sum_{k=1}^{\infty} \frac{(x-5)^k}{3^k k}$ for $x \in I$ what is $f^{(21)}(5)$?

Using Taylor's Theorem: $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$

$$\text{Then } f(x) = \sum_{k=0}^{\infty} \frac{(x-5)^k}{3^k k} = \sum_{k=0}^{\infty} \frac{f^{(k)}(5)}{k!} (x-5)^k$$

$$\text{Then } \frac{f^{(k)}(5)}{k!} = \frac{1}{3^k k}; \text{ Thus, } f^{(21)}(5) = \frac{k!}{3^k k} = \frac{(k-1)!}{3^k}$$

$$= \frac{(21-1)!}{3^{21}} = \frac{20!}{3^{21}}$$

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Problem #4:

Suppose f has derivatives of all orders on $(-1, 1)$ and there exists a constant k such that $|f^{(n)}(x)| \leq n! k$ for all $x \in (-1, 1)$, $\forall n \in \mathbb{N}$.

Prove that the Taylor Series for f centered at 0 converges to f on $(-1, 1)$. Want to show:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x)^n; \text{ Given: } |f^{(n)}(x)| \leq n! k$$

$$\text{So we have } \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x)^n \leq \sum_{n=0}^{\infty} \frac{n! k}{n!} (x)^n$$

true because $|f^{(n)}(x)| \leq n! k$ for any $x \in (-1, 1)$, which contains 0 and thus holds for $|f^{(n)}(0)| \leq n! k$

$\sum_{n=0}^{\infty} \frac{n! k}{n!} (x)^n = \sum_{n=0}^{\infty} k x^n$. This is a geometric series where $a = k$ and $r = x$. Since $x \in (-1, 1)$, then $|r| < 1$, which implies the geometric series converges.

By comparison test, we have shown that the Taylor Series centered at 0 converges to a function f on the interval $(-1, 1)$.